Clock-Sensitive Waits and Fault-Aware Repair in Zephyr RTC Tests

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ABSTRACT

Firmware tests often wait a fixed time for a hardware event. A slower but datasheet-legal clock can make a reference-compatible program fail that test. We study clock-sensitive waits and fault-aware repair using one Zephyr STM32 real-time clock (RTC) driver and the existing Renode model. Three selected upstream cases and three generated suites fail at legal clock settings; four repairs make each pass all 102 tested settings while retaining measured slow-fault detection. Three repairs also make a fourth upstream counter case pass all 102 settings after 44 original failures. An 11-run control verifies modeled 10/32/64 kHz: right-sizing and polling retain this case's slow-fault detection, while naive extension loses it. An older control misses its strict bound by 18 ms. The full counter application retains downstream failures. Right-sizing adds 11 slow-fault detections in different generated counting tests; polling adds none on common-evaluable pairs. All 180 reruns in six fixed-setting cells agree, showing cross-setting sensitivity without observed within-cell flakiness. This is a case study of one shared simulator model, not a developer-test prevalence or general repair rate.

KEYWORDS

embedded testing, timing sensitivity, test repair, RTC, fault detection

1 INTRODUCTION

An RTC test schedules an alarm ten calendar seconds ahead, waits thirteen simulated (Renode virtual) seconds, and asserts that a callback arrived. Its wait assumes that the RTC clock and callback path fit the observation window. At a slower but datasheet-legal clock, the alarm target itself may occur after the assertion. Extending the wait can fix the legal-clock failure. An excessively long deadline can also accept a faulty clock, so a passing repaired test alone is insufficient.

Our question is: **when legal clock variation changes a test verdict, can we repair the timing assumption without weakening measured fault detection?** We analyze paired repair records for one Zephyr STM32 RTC driver in Renode. The original fault-qualified result concerns six subjects in Table 1. A separate full-application counter extension tests a fourth upstream waiting-sensitive case; all repairs fix its legal-clock failures, but naive extension accepts the activated slow fault that right-sizing and polling reject. We also distinguish deadline sizing, state polling and added alarm checks.

Connection to flaky tests. Timing assumptions can make tests unreliable across operating conditions [4, 5, 8]. Here, a consistent pass at one fixed frequency and failure at another is *timing sensitivity*, not observed nondeterminism at one setting. We also ran 30 fresh processes in each of six fixed test-frequency cells: all 180 verdicts agree within their cells. This pilot finds no same-setting flakiness in those cells; it cannot rule out rare flakiness elsewhere. We did not measure a device-frequency population distribution or sample developer tests randomly.

Virtual time (s)	One upstream callback iteration
0	Set RTC time; alarm is ten calendar seconds ahead.
3	Check that no callback has arrived early.
13	Original final check: callback required.
18.824	Ideal target arrival at legal 17 kHz.
20.824	Repaired deadline: require callback by here.

Figure 1: Why the 13-second wait fails. Times illustrate zero initial subsecond phase and omit delivery latency; they are not measured arrival times.

2 APPROACH: REPAIR THE WAIT, RECHECK THE ORACLE

2.1 A clock-derived deadline

The pinned Zephyr v4.3.0 source [9] selects the low-speed internal oscillator (LSI) for the STM32F4 Discovery RTC. DS8626 Rev 12 lists 17-kHz minimum, 32-kHz typical and 47-kHz maximum at 3 V and −40 to 105 °C [7]. We test both endpoints and the nominal typical rate; the datasheet does not give their population probabilities. With a fixed 128×250 prescaler, ten RTC seconds ideally take

$$t(f) = 320{,}000/f \quad \text{simulated seconds, with } f \text{ in Hz.}$$

At 17 kHz this is 18.824 s, beyond the original 13-second wait. Its ideal boundary is 24.615 kHz. These are manually source-derived thresholds, not an automatically inferred specification; full-application phase effects can require a band rather than one boundary.

The repair ceiling is **20.824 simulated seconds**: the ideal slow-clock duration, 18.823529… s, rounded up to the next millisecond (18.824 s), plus **two seconds of unvalidated experimental allowance**. It is not a measured or guaranteed interrupt-to-callback latency bound. Kernel uptime measures the wait while CPU/SysTick remain fixed. Figure 1 makes the original waiting assumption concrete.

In the upstream callback source, lines 120 and 132 sleep for constants of three and ten seconds; lines 126–129 check for premature callbacks and lines 140–141 require arrival. The archived original fails at 17 kHz but passes at 32 kHz; a task-specific 17-kHz reference passes with a longer bounded wait. V1/V1b retain the

early check and sleep only until the revised deadline; V2 can finish when the completion predicate becomes true. At the injected 10-kHz slow fault, the same target ideally takes 32 s, so both the original and repaired deadlines reject it. This explains why extending this subject's wait can preserve its slow-fault check.

2.2 Paired mechanical variants

V0 is the original test. **V1**, naive extension, uses the larger of the original wait and 20.824 s; it never shortens an overlong wait. **V1b**, right-sizing, uses a 20.824-second total ceiling while preserving the original fixed-sleep or polling strategy and cadence. **V2** uses the same ceiling and original completion predicate with 20-ms state polling, latching destructive pending reads. **V3** adds missing checks from the frozen task specification: supported alarm-field readback, callback identity/calendar checks, or counting progress/monotonicity as applicable. This deliberately augments upstream oracles. Original targets, early checks and non-time assertions remain. V1 preserves existing service stages (Sol pending r03 can reach 23.824 s); V1b–V3 share one total ceiling. The dated transformation rules were fixed before repair builds.

The later full-application counter extension has its own frozen deadline: V1 waits 13 s, V1b waits 7.648 s, and V2 polls the original callback count for at most 7.648 s. V3 and calendar-alarm offsets do not apply to this relative-counter case. We compare all three variants with archived V0 verdicts at the same 100 frequencies and 17/47-kHz endpoints; we do not pool them with the 33-subject cohort.

We test faults separately: fast 64-kHz and slow 10-kHz oscillator inputs, a nominal-clock *nonarrival fault* that suppresses alarm delivery or freezes counting reads, and nominal-clock alarm-compare offsets of −1 or +1 RTC second. A table *kill* means an activated fault detected by an explicit selected-case assertion. The out-of-envelope oscillator probes test whether a deadline rejects clearly wrong rates; nonarrival tests missing progress or delivery, and offsets test whether the oracle checks the programmed target. These hand-selected operators diagnose those properties, not the prevalence or importance of field faults. Alarm programming, calendar-target arrival and callback delivery are different properties: readback checks programming; calendar equality also assumes delivery before the calendar advances. We did not vary delivery delay.

3 EXPERIMENTAL SETUP AND SCOPE

Scope. We purposively selected three upstream cases for visible timing windows: alarm callback, alarm pending and calendar counting. We later ran full applications with **10 distinct runnable upstream case names**, including those three, from 19 declared names: five runnable RTC API cases, four counter basic cases and one counter seconds case. Four distinct names have waiting-sensitive failures; we repaired the fourth, counter all-channels, in its full application. These are simulated board configurations, not physical boards. They share **one existing Renode STM32F4 RTC model**, so board labels do not provide independent hardware replication. Counter applications use **non-stock RTC-only configurations**, with the existing counter adapter enabled and other timer counters and the RTC API driver disabled; they do not add a second repaired driver. We retained capability predicates. Neither this selection nor the repair subset estimates developer-test prevalence.

Execution. Renode v1.17.0 is an instruction-level embedded simulator [1]. Each run starts a fresh process with unchanged source/ELF, CPU, SysTick and synchronization quantum within its variant. The frozen schedule has 100 shared integer-frequency settings plus two endpoints within the documented interval. Build, five nominal passes and small calibration precede production; the Linux container was calibrated against local verdicts and trace hashes. We ran TACC Lonestar6 jobs under project allocations TRA25006 and CCR26073, not the course allocation; job records identify each account and queue. A dated extension built three counter-test variants and ran their full applications on the same schedule. A separate diagnostic job replayed selected failures and ran clean-state references and oscillator-progress controls; none replaced a primary verdict. The later bracketed rate check used 11 new processes, not a repeated repair schedule.

F4, F767 and F746 use 17–47 kHz; F767 is explicitly an F4 configuration replica. H753 uses the frozen DS12117 Rev 10 revision-V envelope, 29.760–33.600 kHz at VDD 1.62–3.6 V and junction −40 to 105 °C [13]. F746 counter basic is excluded after an MPU fault, F072 after no test output, and watchdog after no demonstrated reboot. The F072 divider observation remains unconfirmed.

Attribution and missingness. Same-input failure replays, exact source lines and task-specific references distinguish waiting defects, phase-dependent waits, cleanup cascades and unresolved rate contracts. References check premature behavior and progress under their registered ceiling; a passing progress reference cannot override a failed rate reference. Capability skips and missing verdicts remain separate from passes. Infrastructure-interrupted evaluations without a verdict were resumed under a dated amendment, while finalized harness-error outcomes remain missing.

Repair cohort. We fixed the original roster after the initial clock study and before repair builds: three isolated upstream cases plus 30 generated scalar-budget suites (27 counting, three pending), with no generated callback suite. This selection favors identifiable waiting budgets and is not a general repair-success sample. Each variant has 3,300 seeded legal slots and 66 endpoints; alarm offsets apply to five alarm subjects. We pair only subjects evaluable in both variants. We report the later counter extension separately as one additional upstream case in a full application.

4 RESULTS

4.1 Six sensitive subjects: repaired waits retain slow-fault detection

Each of the six subjects passes the tested legal settings after every repair and retains its measured slow-fault detection. Table 1 shows the three upstream subjects and three generated pending suites separately. Every repaired subject passes all 102 legal settings with zero missing verdicts. Naive extension already achieves this result: it does *not* weaken slow-fault detection in these six subjects.

As supporting detail, the six subjects contribute 612 shared subject-setting pairs: V0 produces 500 passes and 112 failures; each repair produces 612 passes. These are **six subjects, not 612**

Table 1: The six originally sensitive subjects. Each repair is compared with the same subject's V0 records. All six detect the slow fault in V0 and each repair, with no missing outcomes.

	V0 failures		Legal passes / 102 settings				Slow fault
Subject	Seeds / 100	Ends / 2	V1	V1b	V2	V3	V0 → each repair
Upstream, isolated cases							
Alarm callback	26/100	1/2	102/102	102/102	102/102	102/102	detect → detect
Alarm pending	26/100	1/2	102/102	102/102	102/102	102/102	detect → detect
Calendar counting	17/100	1/2	102/102	102/102	102/102	102/102	detect → detect
Generated, base prompt							
Sol pending r01	11/100	1/2	102/102	102/102	102/102	102/102	detect → detect
Sol pending r03	13/100	1/2	102/102	102/102	102/102	102/102	detect → detect
Astra pending r05	13/100	1/2	102/102	102/102	102/102	102/102	detect → detect

Table 2: F4 full counter application, 102 paired legal settings per variant. Cells are pass/fail/skip counts for distinct case names; only the all-channels wait was edited.

Variant	All channels	Short relative	Single shot
V0	58/44/0	58/0/44	58/44/0
V1	102/0/0	88/14/0	87/15/0
V1b	102/0/0	88/14/0	87/15/0
V2	102/0/0	88/14/0	87/15/0

Table 3: Full 33-subject cohort. Seed F/M: failures/missing out of 3,300 slots. Fault cells: detections/evaluable. Median is whole-test virtual time.

Variant	Seed F/M	Slow	Early	Late	Median (s)
V0	106/9	21/33	4/5	4/5	11.0
V1	0/7	21/33	4/5	4/5	11.0
V1b	0/5	32/32	4/5	4/5	11.0
V2	0/3	33/33	4/5	4/5	11.0
V3	0/2	33/33	5/5	5/5	11.0

independent tests. Their repeated clock settings describe each subject's sensitivity. The 112 original failing slots have passing task-specific references and become observed passes under all repairs. This result concerns the original isolated contexts, not all expanded case names or untested clock settings.

4.2 Fourth upstream case: legal repair and divergent fault response

The full F4 counter application exposes a distinct waiting defect in `counter_basic.test_all_channels`. The source requests a two-tick relative alarm, waits 1.5 times its nominal conversion (3 s), then requires one callback. That assertion fails at 44 of 102 tested legal frequencies, including 17 kHz, and passes at the nominal 32 kHz and 47 kHz. The driver may add a compensating tick; at 17 kHz, three ticks ideally need 5.647 s before callback delivery. The prespecified 13-s extension, 7.648-s right-sized wait and 7.648-s polling deadline each pass the same 102 settings (Table 2). This is **one additional upstream case**, not 306 independent repairs or a second driver.

The full application does not become failure-free. The original all-channels failure skipped the later short-relative case at 44 settings. Once the repaired wait lets it run, short-relative fails at 14 settings. The later single-shot case fails at 15 settings after repair, down from 44 original failures: 29 original failures disappear on paired runs, while 15 remain. We do not count these as additional repaired tests or label every residual failure a cleanup cascade. Seventeen representative full-application replays match the primary case vectors and traces. Three clean-state functional references pass at 17 legal settings each, covering all 87 residual failing case slots across variants; they do not reproduce the original application's phase and order.

The selected 10-kHz verdict is fail in V0, pass in V1, and fail in V1b/V2; all four 64-kHz selected probes pass, and same-input replays agree. We preregistered a separate rate control that brackets two counter reads with independent SysTick timestamps, roughly ten seconds apart. All five nominal, three slow and three fast fresh processes meet their distinct prespecified tick-count bands: the observed deltas are 10, 3 and 20 ticks, respectively. Under the pinned model, this qualifies the three injected rates and the selected-case slow-fault outcome: **V1 loses detection, while V1b/V2 retain it**. The earlier five-tick progress control still measures 16.040 s at 10 kHz, **18 ms above its preregistered 16.022-s bound**. We preserve that failure; its cause and the exact five-tick timing remain unresolved. The new coarse rate check neither repairs the old bound nor measures callback latency. No new nonarrival or offset probe was qualified for this case.

4.3 A separate result: right-sizing over-padded counting tests

Right-sizing adds 11 slow-fault detections in different, already clock-tolerant generated counting subjects. Across the full 33-subject cohort, V1 retains V0's 21/33 slow-fault detections because it preserves long waits. V1b detects 32/32 evaluable slow faults: the 32 common V1/V1b pairs retain 21 detections, gain 11 and lose none. All 11 gains are generated counting subjects with 102 V0 legal passes and no missing legal verdicts. They correct over-padding; they are *not* the six sensitive subjects in Table 1.

V0 has 106 seeded and six endpoint failures. Every repair has zero observed legal failures across the cohort, but V1/V1b/V2/V3

Table 4: Distinct case names per configuration. Run: ever runnable; Skip: always capability-skipped. Wait, Cascade and Rate are separate failure categories.

Configuration	Run	Skip	Wait	Cascade	Rate
F4	10	9	4	1	1
F767 (F4 replica)	10	9	4	1	1
F746	6	2	3	0	1
H753	10	9	0	0	0

retain respectively 7/5/3/2 missing seeded outcomes; all 66 endpoints are observed passing. There are 29 finalized missing outcomes across legal and fault probes. They are not counted as successes. All 458 primary assertion failures replay exactly, including raw trace hashes.

4.4 Polling and added alarm checks do different work

Polling adds no paired slow-fault detection, while extra alarm checks detect a previously accepted programming error. V2 detects the slow fault in 33/33 subjects, but all 32 common V1b/V2 pairs already detect it under V1b. The remaining V1b outcome is missing, so the denominator change supplies no evidence of a polling gain. Polling instead shortens the upstream callback's median seeded whole-test duration from 42.0 s (V1b) to 20.75 s (V2). The pooled median is 11.0 s for every variant and hides this difference; whole-test time is not callback latency.

V0–V2 detect each alarm-offset sign in 4/5 applicable subjects. V3 reaches 5/5 for each sign by adding detection in the upstream callback. Readback and callback-calendar checks were introduced together; these records cannot isolate timestamp checking's effect. Across adjacent variants, no detected fault becomes accepted on common-evaluable subject pairs. Nonarrival is detected in 32/32 evaluable V0 subjects and 33/33 after repair; its missing V0 verdict is not a paired gain. No evaluable fast-oscillator probe is detected (0/32 for V1b, 0/33 otherwise). Preservation of this measured fault set therefore does not establish a complete oracle.

4.5 Expanded applications: scope and failure attribution

Four waiting-sensitive names do not mean that every failing case is an independent test defect. Table 4 describes the original V0 expansion and separates waiting failures from cleanup cascades and unresolved rates. Across 11 application/configuration combinations, 1,122 launches produce 3,022 passing, 470 failing and 3,138 capability-skipped case slots, with no missing primary case verdicts. Per simulated configuration, waiting-sensitive names are 4/10 runnable names on F4 and its F767 replica, 3/6 on F746, and 0/10 on H753. These are within-cohort fractions, not population prevalence estimates.

Pending has the ideal 24.615-kHz boundary. F4 full-application callback failures span sampled 17.000–23.673 kHz (passes 23.847–47.000); counting failures span 17.000–20.500 (passes 20.701–47.000). The counter all-channels wait has a source-derived phase envelope of 21.333–32.000 kHz. These phase-dependent cases cannot all be summarized by one exact frequency threshold: writing the RTC calendar preserves the subsecond count in the existing model. Relative to isolated runs, F4 full/isolated seeded failure counts are pending 26/26, callback 22/26 and counting 12/17. Phase carryover and ordering are a source-supported explanation, *not directly measured phase.*

A later counter single-shot setup failure follows skipped cleanup and is a cascade, not a fifth independent waiting-sensitive name. Counter seconds fails over sampled 17.000–26.168 kHz and passes 27.086–47.000, but its 105 rate references also fail while progress checks pass. It remains a firmware/simulator/accuracy-contract investigation, not a demonstrated waiting defect. All 470 failing slots have reference outcomes: 277 waiting references pass, 88 clean-state cascade references pass and 105 rate references fail. H753 has no observed failures in its narrower modeled interval; one counter case skips nominally but runs at some settings. This contrast varies the range/configuration within the shared model, not silicon validation. Source excerpts and complete per-case bands remain in the artifact.

4.6 Secondary generated-test context

The generated cohort covers only the original three task purposes: `gpt-6-luna`, `gpt-6-sol` and `gpt-6-astra` each produced five source-only ztests (Zephyr's test framework) per task and prompt. Of 90 attempts, 66 pass functional/build/five-nominal gates. The base prompt omits numeric clock ranges; the other adds ranges and a delivery reminder. Suite-level clock sensitivity is 3/34 base versus 0/32 with ranges (two-sided Fisher $p = 0.239$); 12 original seeded verdicts remain missing. Models received purpose-derived specifications, not upstream code or execution feedback. The specifications requested stronger observables, including configured-time callbacks/IDs and pending readback/not-before checks. We report executable/nominal yield and seeded-fault detection as limited usefulness proxies; neither measures maintainer value. With no same-spec human baseline, neither oracle differences nor this small prompt contrast establishes model superiority or an isolated numeric-range effect. Detailed generation, yield, two-axis, fault and distribution tables are retained in the artifact rather than expanding this comparison here.

Fixed-setting repeatability. We reran unchanged full applications in 30 fresh processes per cell. The F4 callback fails 30/30 at 23.673 kHz and passes 30/30 at 23.847 and 32 kHz. F4 counter all-channels fails 30/30 at 29.143 kHz and passes 30/30 at 29.352 and 32 kHz. All 180 outcomes are explicit and agree with archived same-frequency vectors. This pilot shows no mixed verdict within these six cells; it does not establish a population flakiness rate or rule out rare nondeterminism.

5 DISCUSSION AND LIMITATIONS

What the repair evidence supports. The original six-subject result shows that those waits accept tested legal clocks while retaining measured slow-fault detection. The fourth upstream counter case adds legal-clock repair evidence in a full application, exposes residual failures, and shows a qualified selected-case slow-fault

loss under naive extension but not right-sizing/polling. The different 11-subject counting result shows why preserving an already overlong wait can leave a weak slow-clock oracle. Right-sizing supplies those gains without polling; polling reduces time; augmented checks improve alarm-offset detection. None establishes semantic equivalence for every defect or an automatic general repair procedure.

Clock ranges and distributions. The pinned HSI (high-speed internal oscillator) context study varied 15.84–16.16 MHz (±1% at 25 °C), with integer 99–101-MIPS CPU mapping: no failures among 943 upstream and 3,270 generated evaluable seeded runs; 87 seeded outcomes were not evaluable. Different tasks, the narrow range and throughput approximation prevent a direct HSI-versus-LSI causal comparison. The RTC study isolates LSI variation with fixed CPU/SysTick; its 17–47-kHz range is much wider. The HSI result does not imply HSI can never matter.

Neither uniform sampling nor a datasheet interval measures how frequencies are distributed across devices. We lead with boundaries/bands, retaining uniform and legal-conditioned normal sensitivity calculations in the artifact as illustrative assumptions only. ST documents describe oscillator accuracy/calibration, not a usable F407 population distribution for this analysis [10–12]. The board survey's 106 LSI versus 40 LSE (external-crystal) RTC-enabled targets supplies context, not a flaky-test prevalence estimate.

Practical guidance and limits. Derive waits from the worst-case legal clock, justify a delivery allowance, retain early-arrival checks against an independent time reference, and assert the RTC calendar/programming target rather than eventual arrival alone. Our extra two seconds is an unvalidated experimental allowance: no explicit interrupt-to-callback latency bound was found in 33 scanned files. The earlier counter control started uptime before its initial read and polled every 20 ms; saved traces sample only every 0.5 s. These facts do not explain its 18-ms discrepancy. The new read-bracketing control checks coarse rate, not that strict elapsed-time bound or callback latency. Fast-fault blindness, purposive selection, shared model behavior and non-stock counters limit external validity. One unblinded reviewer coded source/padding; related suites and frequencies do not replicate. The artifact's 2× cutoff is descriptive. Missing verdicts remain missing.

6 RELATED WORK

Flaky-test studies report asynchronous waits and environment assumptions [4, 8]; FlakeRake perturbs timing to reproduce intermittent failures [5]. Our clock changes expose sensitivity across settings, while the six fixed-setting pilot cells show no within-cell mixed verdicts. Interrupt-driven testing likewise explores scheduling conditions [6]. Mutation testing and oracle research motivate fault checks after a repair, rather than accepting nominal passes alone [2, 3]. Our contribution is the paired separation of wait repair, deadline sizing and oracle augmentation in this RTC setup, including a counter case where naive extension accepts an activated slow fault.

7 CONCLUSION

Each of six original sensitive subjects passes its tested legal settings after repair and retains measured slow-fault detection. A fourth upstream counter case passes all 102 tested legal settings after repair, but naive extension accepts its activated slow fault while right-sizing and polling reject it; downstream failures and an earlier strict control discrepancy remain. Separately, right-sizing improves detection in 11 already clock-tolerant counting subjects. These results support targeted clock-sensitive wait repair with explicit limits on faults and the shared simulator model.

AI USE

LLMs generated the evaluated suites. Automated tooling assisted execution, auditing and manuscript preparation under author direction; the authors review the evidence and take responsibility for the claims.

CODE AND DATA AVAILABILITY

Code, frozen tests, amendments and data dictionary; results. The unchanged evidence release v1 contains an *older PDF*. The present PDF, TeX, counter records, bracketed control and claim audit are versioned as manuscript v4. `make course-tables` checks the pinned archive and tables without Renode; the v4 supplement includes the raw control job archive.

REFERENCES

[1] Antmicro. n.d. Renode time framework documentation. Renode documentation (accessed 2026-11-04). Renode docs

[2] Earl T. Barr, Mark Harman, Phil McMinn, Muzammil Shahbaz, and Shin Yoo. 2015. The Oracle Problem in Software Testing: A Survey. *IEEE Transactions on Software Engineering* 41, 5, 507–525. DOI

[3] Yue Jia and Mark Harman. 2011. An Analysis and Survey of the Development of Mutation Testing. *IEEE Transactions on Software Engineering* 37, 5, 649–678. DOI

[4] Qingzhou Luo, Farah Hariri, Lamyaa Eloussi, and Darko Marinov. 2014. An Empirical Analysis of Flaky Tests. In *Proceedings of the 22nd ACM SIGSOFT International Symposium on Foundations of Software Engineering*, 643–653. DOI

[5] Shanto Rahman, Aaron Massey, Wing Lam, August Shi, and Jonathan Bell. 2024. Automatically Reproducing Timing-Dependent Flaky-Test Failures. In *2024 IEEE Conference on Software Testing, Verification and Validation (ICST)*, 269–280. DOI

[6] John Regehr. 2005. Random Testing of Interrupt-Driven Software. In *Proceedings of the 5th ACM International Conference on Embedded Software*, 290–298. DOI

[7] STMicroelectronics. 2026. STM32F405xx, STM32F407xx Datasheet. STMicroelectronics datasheet DS8626 Rev 12. ST DS8626

[8] P. E. Strandberg, T. J. Ostrand, E. J. Weyuker, W. Afzal, and D. Sundmark. 2020. Intermittently Failing Tests in the Embedded Systems Domain. In *Proceedings of the 29th ACM SIGSOFT International Symposium on Software Testing and Analysis*, 337–348. DOI

[9] Zephyr Project. 2025. Zephyr RTOS v4.3.0 (release commit 3568e1b6d5cd). Zephyr Project source revision. Zephyr commit

[10] STMicroelectronics. 2018. How to optimize STM32 MCUs internal RC oscillator accuracy. Application note AN5067, Rev 1. ST AN5067

[11] STMicroelectronics. 2023. How to calibrate internal oscillators on STM32G0 MCUs. Application note AN5126, Rev 2. ST AN5126

[12] STMicroelectronics. 2024. STM32F4 reference manual RM0090, Rev 21, Sec. 6.2.11. ST RM0090

[13] STMicroelectronics. 2026. STM32H753xI datasheet DS12117, Rev 10, Tables 38 and 135 (revision V). ST DS12117